

WELCOME TO

Japan–Singapore Workshop on Applied Quantum Optimization

~ Quantum Infinity for You (QI4U) ~

25–27 August 2026

Singapore

Optional Day 4 • Follow-up

Three days of **keynotes, hands-on coding and team projects** spanning gate-based quantum computing and quantum annealing — built and taught together by **SMU** and **Tohoku University**. Come with a problem; leave with a formulation, a working circuit and a team.

THE THREE-DAY ARC

DAY 1

Foundations, Applications and Challenges

DAY 2

Hands-on Development and Group Work

DAY 3

Project Presentations and Future Directions

KEYNOTE LECTURES



Prof. Masayuki Ohzeki

Tohoku University

Introduction to Quantum Computing and Our Challenges



Prof. Hoong Chuin Lau

Singapore Management University

Hybrid Quantum-Classical Approaches for Optimization

WHAT YOU WILL ACTUALLY DO

● Gate-based QC

- › QAOA and VQE on IBM Quantum hardware
- › Custom Penalty VQE for constrained optimization
- › Research showcases from the SMU group

● Quantum Annealing

- › How to formulate QUBO problems
- › Traffic and traffic-light optimization
- › Portfolio optimization

● Group Project

- › Problem exploration and idea development
- › Formulation, implementation, evaluation
- › Final group presentations on Day 3

DAY 4 OPTIONAL

Follow-up Research Collaboration Session. An optional follow-up session may be organized for participants interested in continuing project discussions, preparing joint research proposals, or exploring longer-term academic and industry partnerships.

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Programme

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Applied Quantum Optimization
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Day 1 Foundations, Applications and Challenges

09:30–10:00	Registration
10:00–10:15	Welcome & Opening Remarks Prof. Hoong Chuin Lau, SMU
10:15–11:00	Keynote 1 • Prof. Masayuki Ohzeki, Tohoku University Introduction to Quantum Computing and Our Challenges
11:00–11:45	Keynote 2 • Prof. Hoong Chuin Lau, SMU Hybrid Quantum-Classical Approaches for Optimization
11:45–13:00	Lunch
13:00–13:30	Research Showcase I • Xinwei Lee, SMU Solving Constrained Optimization with Custom Penalty VQE
13:30–14:00	Research Showcase II • Monit Sharma, SMU Newsvendor Optimization by Quantum Tail Estimation
14:00–14:45	Gate-based QC I • Monit Sharma, SMU Implementing QAOA and VQE on IBM Quantum Hardware

14:00–14:45	Gate-based QC II • Xinwei Lee, SMU Implementing Custom Penalty VQE for Constrained Optimization
14:45–15:00	Workshop Specials • Takuma Yoshihara, Tohoku University Introduction to Prof. Ohzeki AI Companion
15:00–15:30	Group Photo and Coffee Break
15:30–16:00	Quantum Annealing I • Mizuki Muto, Tohoku University Applications of Quantum Annealing and How to Formulate QUBO Problems
16:00–16:30	Quantum Annealing II • Takumi Okuyama, Tohoku University Optimizing Traffic and Traffic Lights using Quantum Annealing
16:30–17:00	Quantum Annealing III • Kouki Fujiwara, Tohoku University Portfolio Optimization using Quantum Annealing
17:00–18:00	Grouping and Discussion on Group Member Assignments

■ Keynote ■ Hands-on tutorial ■ Quantum annealing ■ Break

Day 2

Hands-on Development and Group Work

10:00–12:00	Group Work I Problem Exploration and Idea Development
12:00–13:00	Lunch
13:00–15:00	Group Work II Topic Selection, Formulation, Implementation, and Experimental Evaluation
15:00–15:30	Coffee Break
15:30–18:00	Group Work II Topic Selection, Formulation, Implementation, and Experimental Evaluation

Day 3

Project Presentations and Future Directions

10:00–12:00	Group Work III Final Computation, Analysis, and Presentation Preparation
12:00–13:00	Lunch
13:00–15:00	Final Group Presentations
15:00–15:30	Coffee Break
15:30–17:00	Final Group Presentations
17:00–18:00	Closing Remarks and Discussion on Future Collaborations

Resources — scan to join



Discord server
discord.gg/5GrCh7Pn9



GitHub
github.com/SMU-Quantum/
sg-jp-workshop

Administrative info

Wi-Fi	eduroam
Lunch	Self-paid Venues announced on site
Coffee breaks	Provided
Slides & notebooks	GitHub repository SMU-Quantum / sg-jp-workshop

Questions during the workshop? Ask on **Discord**.

